

Understanding the Transition from Memorization to Generalization in Diffusion Models for Cosmology

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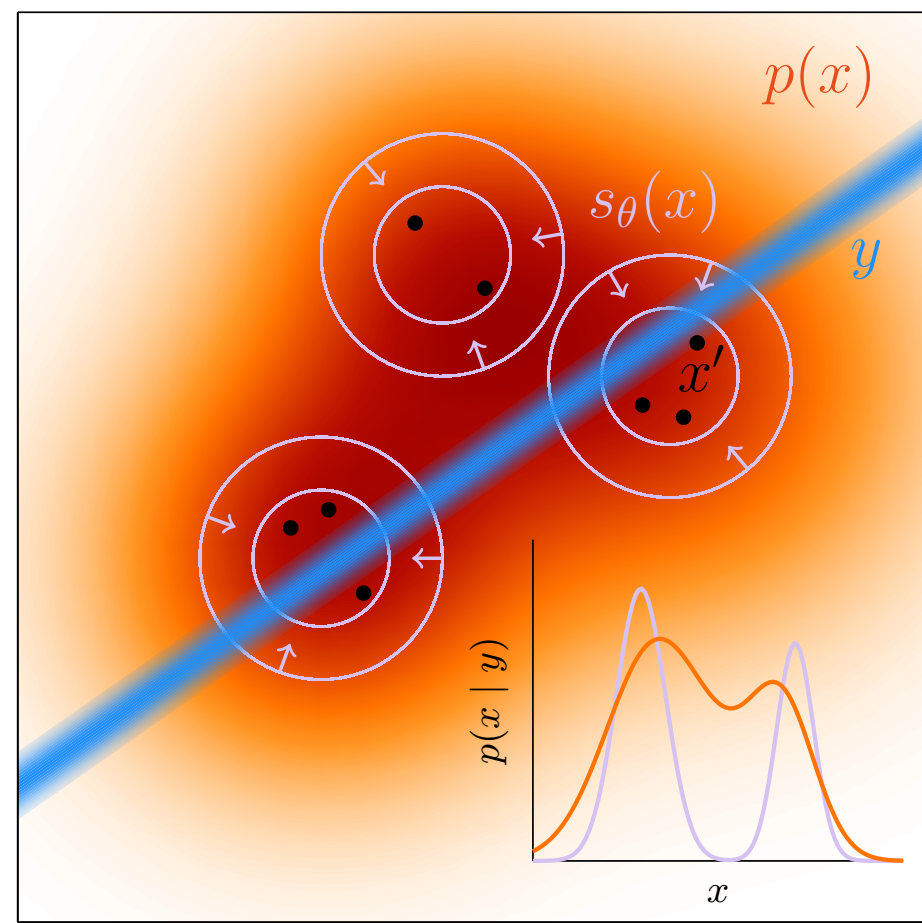
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The question

For inference, a diffusion generator becomes a prior over HI fields x . The cartoon shows the failure mode: orange is the true field distribution $p(x)$, black dots are training maps, purple $s_\theta(x)$ is the learned diffusion score/prior, and blue is the data constraint y . If $s_\theta(x)$ wraps only training islands, $p(x|y)$ can be sharp but biased. We therefore test:

- Novelty:** when do generated maps stop matching individual training slices?
- Conditioning:** does changing θ_{ask} change the recovered cosmology?
- Fidelity:** do novel maps retain the one-point PDF and power spectrum?



Data & models

- **CAMELS** (IllustrisTNG) LH set: 1000 simulations → 128 slices each → 128,000 candidate 128 × 128 HI maps.
- Six-parameter conditioning: $\theta = (\Omega_m, \sigma_8, A_{\text{SN}1}, A_{\text{AGN}1}, A_{\text{SN}2}, A_{\text{AGN}2})$.
- **Diffusion objective.** We corrupt a real map with noise and train a UNet to predict the velocity target at every noise level, with the cosmology θ as conditioning.

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{x_0, \epsilon, t} \|f_\theta(x_t, t, \theta) - \text{target}\|_2^2.$$

Here the config uses v_prediction: target = $v_t = \sqrt{\alpha_t} \epsilon - \sqrt{1 - \alpha_t} x_0$. Ho et al. 2020 (DDPM).

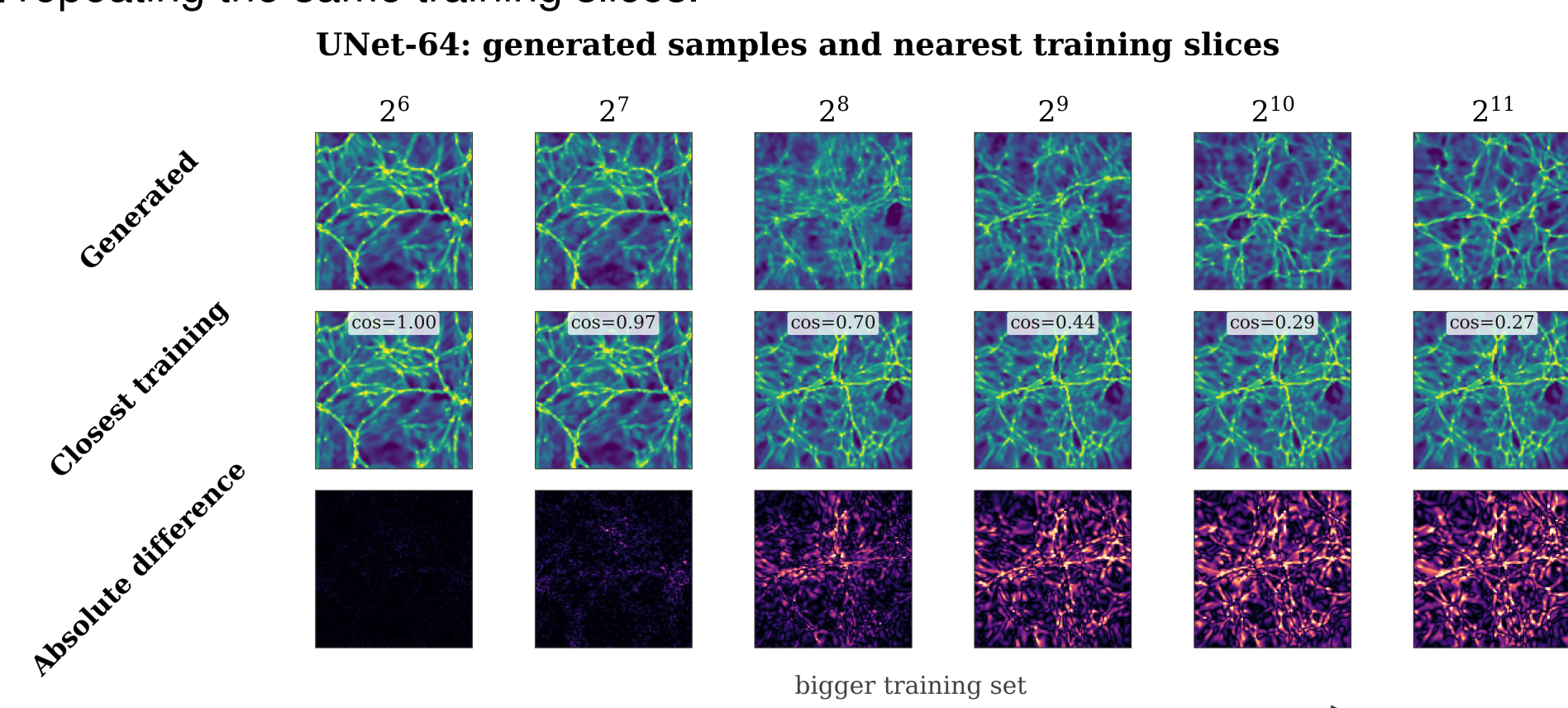
- **Sampling:** 50-step DPM-Solver, 8× faster than the 500-step DDPM baseline at matched quality.
- **Training-set sweep:** $N_{2D} = 2^6 \rightarrow 2^{15}$ fields, matched optimizer-update budget; runs on U-M Great Lakes HPC.

Detecting memorization

For each generated map x_j , compute its similarity to every training map y_i and keep the closest match:

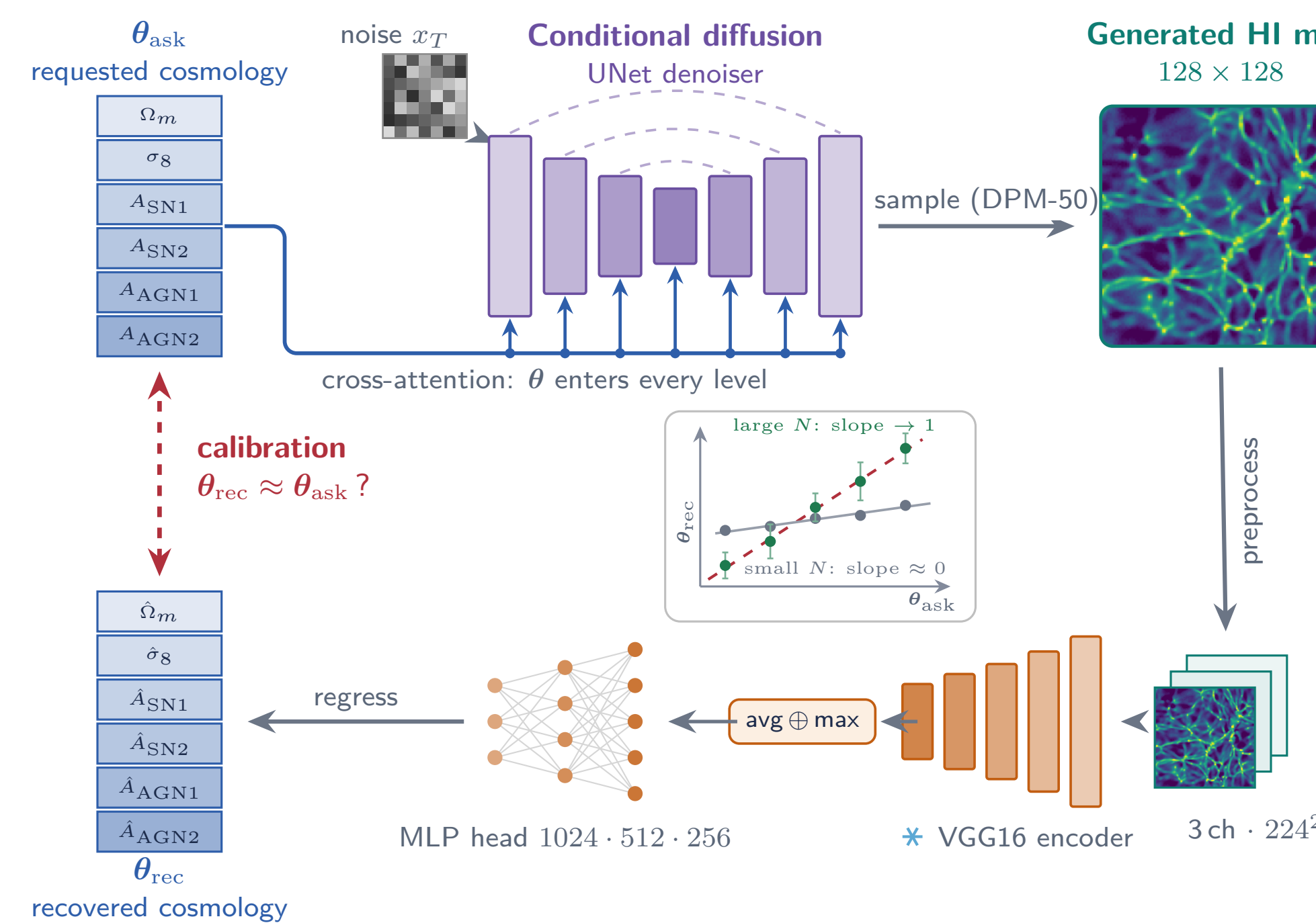
$$s_j = \max_i \text{sim}(x_j, y_i)$$

A high s_j means x_j is nearly a copy of one training map. We measure sim in pixel, PCA, and SSCD space. Nearest-neighbor plots test individual copying; reproducibility plots compare the generated distributions from independently trained runs. Good runs can match statistically without repeating the same training slices.



Calibration: does the generator obey θ ?

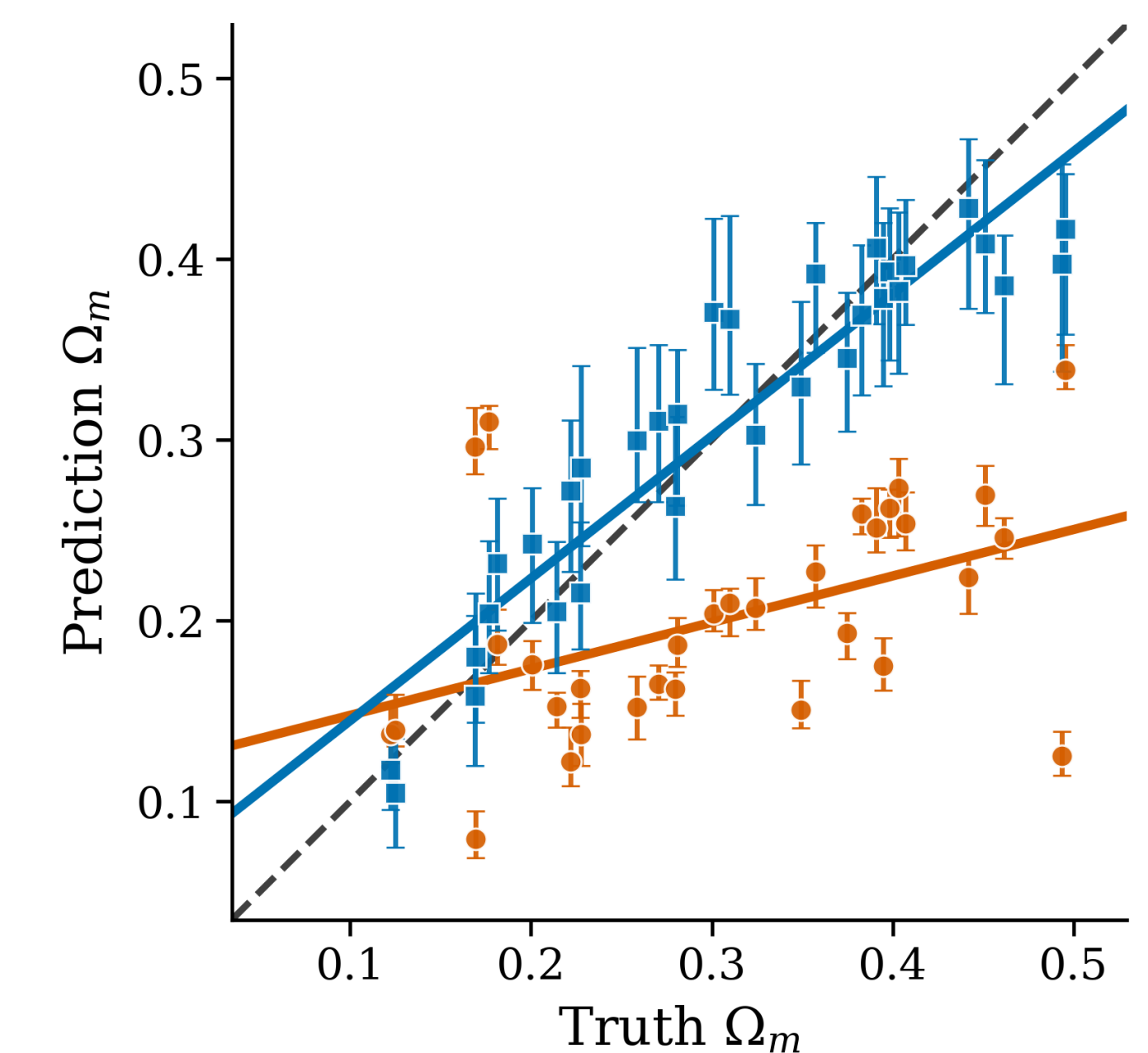
Left to right, a cosmology vector θ_{ask} conditions the UNet through cross-attention while it denoises random noise into an HI map. The map is copied to 3 channels, resized, and encoded by frozen VGG16 features; an MLP head returns θ_{rec} . The dashed loop asks whether $\theta_{\text{rec}} \approx \theta_{\text{ask}}$.



Calibration results

Generated HI fields recover Ω_m

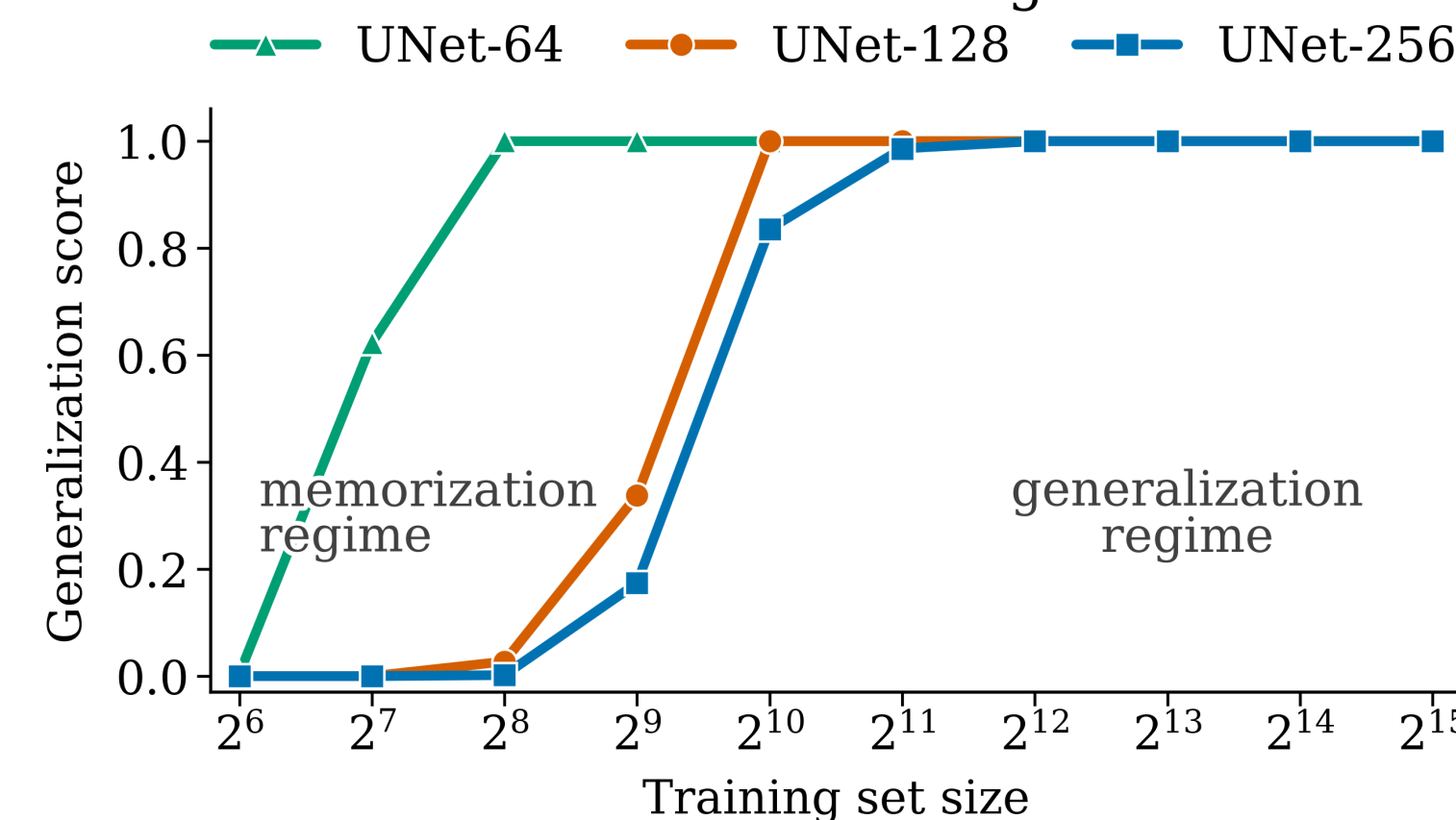
— ideal recovery ● mem. regime ($N_{2D} = 2^7$), slope=0.26 ■ gen. regime ($N_{2D} = 2^{14}$), slope=0.79



Main result: the bars show 16–84% seed scatter at fixed θ_{ask} . In the high-data regime, the medians usually bracket the requested Ω_m within that scatter. In the memorization regime, seeds can return similar training-like maps, so the bars look tight while the median is biased. The slope (0.26 vs 0.79) supports the same point, but uncertainty calibration is the main message.

Memorization → generalization transition

Generated fields become less training-set-like with more data



The generalization score is the fraction of generated maps that are *not* close to any training map,

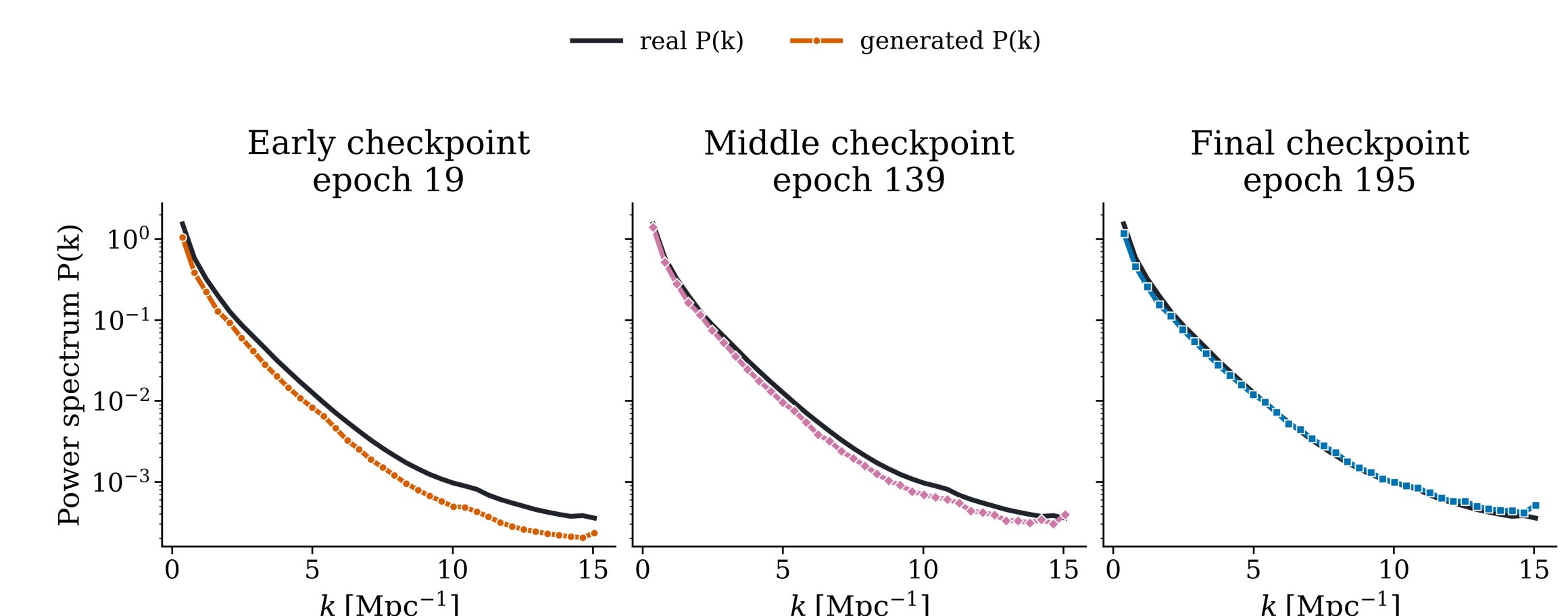
$$\text{GL} = \Pr[s_j < \tau],$$

with τ the 95th percentile of train-vs-train closest-match similarity (PCA space).

- $\text{GL} \approx 0$: almost every generated map sits on top of a training map. The model is **copying**.
- $\text{GL} \approx 1$: generated maps are no closer to training than training maps are to each other; the model makes **new fields**.
- The jump happens over a narrow range of N_{2D} ; in these runs the UNet-256 transition appears at larger N_{2D} (Zhang et al. 2024; Kadkhodaie et al. 2024; Yoon et al. 2023).

Physical fidelity

Generated power spectra approach the real reference during training



Across a single run, generated power spectra move toward the real reference from early to final checkpoints. Early checkpoints suppress power across much of k [Mpc^{−1}]; by the final checkpoint the generated $P(k)$ follows the real shape much more closely.

Takeaways

- The transition appears in cosmological data.** The memorization-to-generalization transition known from natural images also occurs in CAMELS HI maps.
- It explains the error bars.** Small training sets give near-copies: inference looks precise but is biased toward the training values. Past the transition, θ follows the input with honest scatter.
- It gives a practical check.** Nearest-neighbor similarity and encoder calibration reveal an emulator's regime before inference.